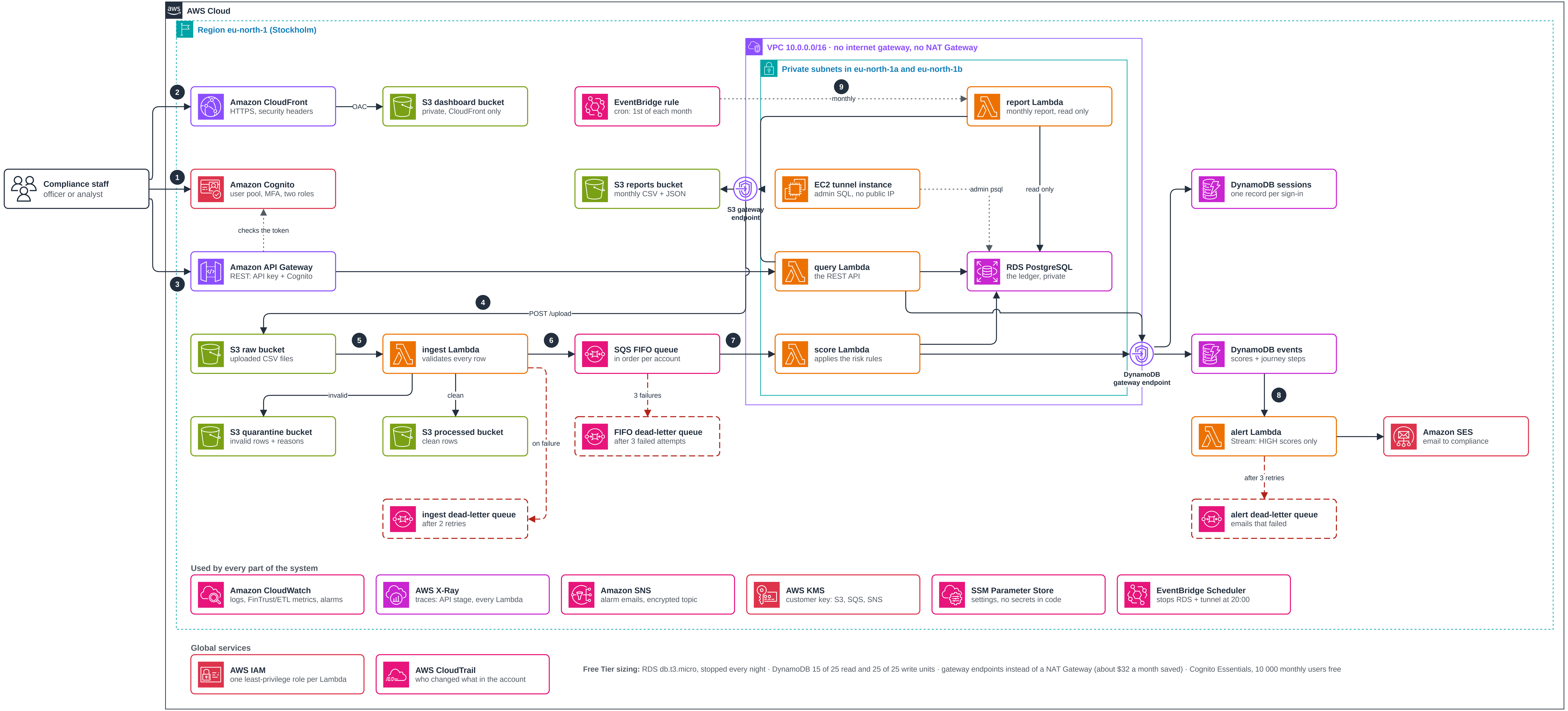
FinTrust Transaction Insights and Risk Monitoring System: AWS architecture

Region eu-north-1 (Stockholm) · serverless and event-driven, within the AWS Free Tier · the numbers follow one transaction and are explained on the right



AWS Cloud

Region eu-north-1 (Stockholm)

VPC 10.0.0.0/16 · no internet gateway, no NAT Gateway

Private subnets in eu-north-1a and eu-north-1b

Compliance staff
officer or analyst

Amazon CloudFront
HTTPS, security headers

S3 dashboard bucket
private, CloudFront only

EventBridge rule
cron: 1st of each month

report Lambda
monthly report, read only

Amazon Cognito
user pool, MFA, two roles

S3 reports bucket
monthly CSV + JSON

S3 gateway endpoint

EC2 tunnel instance
admin SQL, no public IP

DynamoDB sessions
one record per sign-in

Amazon API Gateway
REST: API key + Cognito

query Lambda
the REST API

RDS PostgreSQL
the ledger, private

S3 raw bucket
uploaded CSV files

ingest Lambda
validates every row

SQS FIFO queue
in order per account

score Lambda
applies the risk rules

DynamoDB gateway endpoint

DynamoDB events
scores + journey steps

S3 quarantine bucket
invalid rows + reasons

S3 processed bucket
clean rows

FIFO dead-letter queue
after 3 failed attempts

alert Lambda
Stream: HIGH scores only

Amazon SES
email to compliance

ingest dead-letter queue
after 2 retries

alert dead-letter queue
emails that failed

Used by every part of the system

Amazon CloudWatch
logs, FinTrust/ETL metrics, alarms

AWS X-Ray
traces: API stage, every Lambda

Amazon SNS
alarm emails, encrypted topic

AWS KMS
customer key: S3, SQS, SNS

SSM Parameter Store
settings, no secrets in code

EventBridge Scheduler
stops RDS + tunnel at 20:00

Global services

AWS IAM
one least-privilege role per Lambda

AWS CloudTrail
who changed what in the account

Free Tier sizing: RDS db.t3.micro, stopped every night · DynamoDB 15 of 25 read and 25 of 25 write units · gateway endpoints instead of a NAT Gateway (about $32 a month saved) · Cognito Essentials, 10 000 monthly users free

How one transaction travels

1. Staff sign in with a password and an authenticator code. Cognito issues an ID token for 30 minutes.
2. The dashboard loads from CloudFront; its bucket stays private.
3. Every API call carries the API key (which application) and the ID token (which person). The query Lambda checks the session and the role.
4. POST /upload stores the CSV in the raw bucket, through the S3 gateway endpoint.
5. The S3 event starts the ingest Lambda. Invalid rows go to quarantine with the reason.
6. One message per valid transaction, grouped by account so its order is kept.
7. The score Lambda applies the rules and writes the ledger in one database transaction, then the score and its rule breakdown to DynamoDB.
8. The DynamoDB Stream passes HIGH scores to the alert Lambda, which emails compliance through SES.
9. On the 1st of each month EventBridge runs the report Lambda; CSV and JSON go to the reports bucket.

Arrows

──▶	data or a request
- - -▶	failure path: nothing is lost
·······▶	check, schedule or admin access
dead-letter queue	keeps what failed until it is fixed

Security and network

- No NAT Gateway and no internet route. The three VPC Lambdas reach S3 and DynamoDB through free gateway endpoints, and log in to PostgreSQL with IAM database authentication: no database password exists.

- Security groups: the Lambda group has no inbound rules; the RDS group accepts port 5432 only from the Lambda group and the tunnel instance.

- Journey steps (received, validated, queued, scored, alert sent, and every failure) are written to the events table by the ingest, score and alert Lambdas.